

TIØ4145 Corporate Finance

Exercise 10: Real Options

Deadline: Monday November 10th, 2023, 23:59 – Grading: Passed/Failed

Your answers should be submitted as a PDF file to BlackBoard before the deadline, either written in MS Word, LaTeX or handwritten with a high-quality scan. You are also welcome to upload additional Excel computations if you find it necessary. Guidance to this exercise is given on Thursday November 6th, 8:15 – 10:00 in A32. Solutions to the problems will be posted on BlackBoard after the deadline.

Relevant Curriculum

- Lecture notes from lecture given on November 3rd.
- Berk and DeMarzo, Corporate Finance 5th edition: Chapter 22.

Tasks

Chapter 22 – Real Options

1. You are trying to decide whether to make an investment of \$500 million in a new technology to produce Everlasting Gobstoppers. There is a 60% chance that the market for these candies will produce profits of \$100 million annually, a 20% chance the market will produce profits of \$50 million, and a 20% chance that there will be no profits. The size of the market will become clear one year from now. Currently, the cost of capital of the project is 11% per year. There is a 20% chance that the cost of capital will drop to 9% in a year and stay at that level forever, and an 80% chance that it will stay at 11% forever. Movements in the cost of capital are unrelated to the size of the candy market. Construct the decision tree that shows the choices that you have; to make the investment either today or one year from now.
2. You are a financial analyst at Global Conglomerate and are considering entering the shoe business. You believe that you have a very narrow window for entering this market. Because of Christmas demand, the time is right today and you believe that exactly a year from now would also be a good opportunity. Other than these two

windows, you do not think another opportunity will exist to break into this business. It will cost you \$35 million to enter the market. Because other shoe manufacturers exist and are public companies, you can construct a perfectly comparable company. Hence, you have decided to use the Black-Scholes formula to decide when and if you should enter the shoe business. Your analysis implies that the current value of an operating shoe company is \$40 million and it has a beta of 1. However, the flow of customers is uncertain, so the value of the company is volatile—your analysis indicates that the volatility is 25% per year. Fifteen percent of the value of the company is attributable to the value of the free cash flows (cash available to you to spend how you wish) expected in the first year. If the one-year risk-free rate of interest is 4%:

- a. Should Global enter this business and, if so, when?
- b. How will the decision change if the current value of a shoe company is \$36 million instead of \$40 million?
- c. Plot the value of your investment opportunity as a function of the current value of a shoe company.
- d. Plot the beta of the investment opportunity as a function of the current value of a shoe company.

3. [Voluntary]

It is the beginning of September and you have been offered the following deal to go heli-skiing. If you pick the first week in January and pay for your vacation now, you can get a week of heliskiing for \$2500. However, if you cannot ski because helicopters cannot fly due to bad weather, there is no snow, or you get sick, you do not get a refund. There is a 40% probability that you will not be able to ski. If you wait until the last minute and go only if you know that the conditions are perfect and you are healthy, the vacation will cost \$4000. You estimate that the pleasure you get from heli-skiing is worth \$6000 per week to you (if you had to pay any more than that, you would choose not to go). If your cost of capital is 8% per year, should you book ahead or wait?

4. You own a wholesale plumbing supply store. The store currently generates revenues of \$1 million per year. Next year, revenues will either decrease by 10% or increase by 5%, with equal probability, and then stay at that level as long as you operate the store. You own the store outright. Other costs run \$900,000 per year. There are no costs for shutting down; in that case, you can always sell the store for \$500,000. What is the business worth today if the cost of capital is fixed at 10%?

5. Describe the benefits and costs of delaying an investment opportunity.

6. Genenco is developing a new drug that will slow the aging process. In order to succeed, two breakthroughs are needed, one to increase the potency of the drug, and the second to eliminate toxic side effects. Research to improve the drug's potency is expected to require an upfront investment of \$10 million and take 2 years; the potency research has a 5% chance of being successful. Reducing the drug's toxicity will require a \$30 million up-front investment, take 4 years, and has a 20% chance of success. If both efforts are successful, Genenco can sell the patent for the drug to a major drug company for \$2 billion. All risk is idiosyncratic, and the risk-free rate is 6%.
 - a. What is the NPV of launching both research efforts simultaneously?

 - b. What is the optimal order to stage the investments?

 - c. What is the NPV with the optimal staging?

7. Your firm is thinking of expanding. If you invest today, the expansion will generate \$10 million in FCF at the end of the year, and will have a continuation value of either \$150 million (if the economy improves) or \$50 million (if the economy does not improve). If you wait until next year to invest, you will lose the opportunity to make \$10 million in FCF, but you will know the continuation value of the investment in the following year (that is, in a year from now, you will know what the investment continuation value will be in the following year). Suppose the risk-free rate is 5%, and the risk-neutral probability that the economy improves is 45%. Assume the cost of expanding is the same this year or next year.
 - a. If the cost of expanding is \$80 million, should you do so today, or wait until next year to decide?

 - b. At what cost of expanding would there be no difference between expanding now and waiting? To what profitability index does this correspond?

Exam Question – 2014

You have an idea for a revolutionary consumer electronics product that will require an upfront R&D investment of \$5 million. Initial research will take one year and if things go well, there will be a 75% chance you will develop a successful prototype. If things go badly, there will only be a 25% chance you will develop a successful prototype. It is equally likely that things go well or go badly, so initially, all you know is that there is a 50% chance the R&D will yield a successful prototype. The risk of the R&D process in this period is completely idiosyncratic.

If you do develop a successful prototype, you will need to invest \$40 million at $t = 1$ into manufacturing capacity. Production will then start at $t = 2$; with estimated net cash flows starting at \$3 million and growing at 4% per year.

The risk-free rate is 4%; the market risk premium is 5%; the beta for the net cash flows generated by this type of consumer electronics product is 1.2.

- a. Conditional on developing a successful prototype, what is the NPV at $t = 1$ of investing in the manufacturing capacity to produce this product?
- b. What is the NPV at $t=0$ of the R&D investment?
- c. Suppose that before investing in R&D, you can hire a consulting firm which will be able to tell you for sure whether the initial R&D investment has a 25% or 75% success probability. How much are you willing to pay, if anything, to this consulting firm?

Exam Question – 2018 [Voluntary]

Consider a stock with a current price of 400 NOK. In each of the next two years (2019 and 2020) the stock price will either increase by 25% or decrease by 20%. The stock pays no dividends, and the risk-free interest rate is equal to 7%. Use a two-period binomial model to answer the following questions.

- a. Calculate the price of a European call option with a strike price of 375 NOK.

Assume now that the stock pays a 25% dividend in period 2019.

- b. Recalculate the price of the European option in a) given the new assumption regarding dividends (hint: you need to calculate the option values from 2019 based on ex dividend price, whereas the option value in 2018 is based on the cum dividend price).
- c. Compare the prices in a) and b) and explain the difference (or absence of difference).
- d. What is the price of the American option on the same stock as in b) with the same strike price of 375 NOK? (the same hint as in b) applies)
- e. Compare the prices of the American and the European options in b) and d) and explain the difference (or the absence of difference).

Note: You can choose between replicating portfolio method or risk-neutral probabilities to price the options.

References

Berk, J. and DeMarzo, P. (2017). Corporate Finance, Fifth Edition, Pearson Education, Essex, England.

